PROJECT AND TEAM INFORMATION

Project Title

Skill Bridge : Smart Career & Skill Gap Analysis System

Student/TeamInformation

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PROPOSAL DESCRIPTION(10pts)

Project Overview

Project idea Skill Bridge is proposed as a Java-based application to help students compare their current skills with the skills required for a selected career. Its goal is to display a skill-match percentage, point to the missing or weaker skills, and suggest ways to improve. The problem being addressed. Career requirements, skill assessments and learning resources are often scattered on different websites. Skill Bridge is designed to collect the basic comparison in one step so that students can see the gap between their current capabilities and a target role.

Proposed system workflow Student enters skills and proficiency → selects a target career → client sends a request to the Java server → server validates the request and compares it with career data in MySQL → system calculates match and gap → results and recommendations are returned to the client. Scope boundary The first version concentrates on the basic skill-gap analysis and recommendations only. It is not supposed to provide any guarantees for employment, give m complete career counselling, or act as a job-placement service.. Results depend on the career and skill data maintained by the project team.

Updated Project Approach and Architecture

The proposed approach consists of the following steps:

1. User Input and Career Selection: The student enters their current skills, specifies their proficiency levels, and selects a desired career from the available career categories.

2. Client-Server Communication: The client sends the student's information to the Java server through a socket connection. The server accepts the request and verifies the submitted data before processing it.

3. Database Management: The server extracts the required skills and their expected proficiency levels for the selected career from the MySQL database. The database will contain career categories and their corresponding skill requirements.

4. Skill Gap Analysis: The system compares the student's existing skills with the skills required for the selected career. It identifies matching skills, missing skills, and areas where proficiency needs improvement.

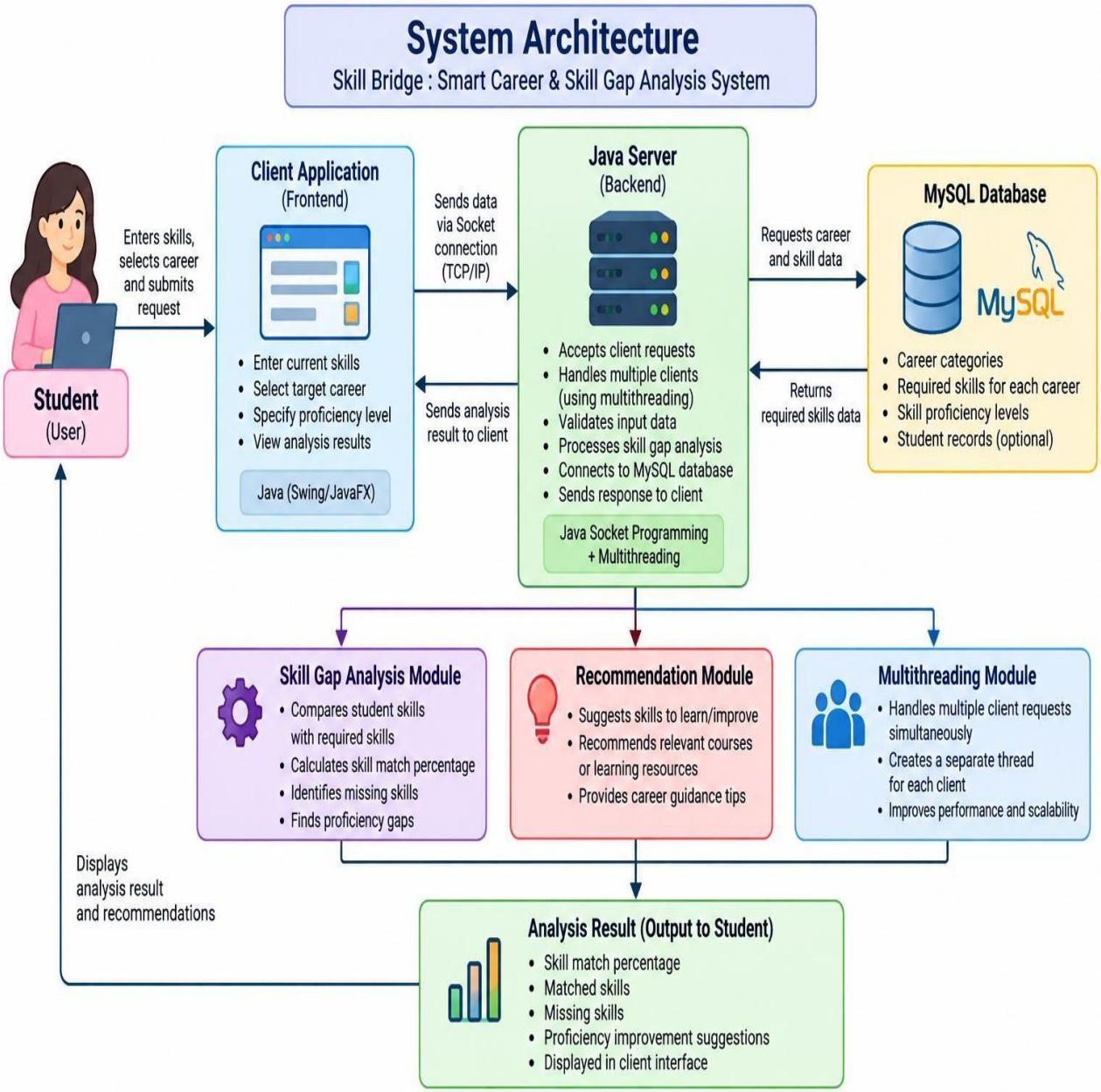
5. Skill Match Calculation: The system calculates a skill-match percentage based on the comparison between the student's skills and the selected career's requirements. The calculation method will be defined and tested to ensure consistent results.

6. Personalised Recommendations: Based on the identified skill gaps, the system generates suggestions that help students determine which skills they should improve or learn to prepare for their desired career.

7. Multithreading and Request Handling: Java multi-threading will be used to process requests from multiple clients , helping the server handle simultaneous users efficiently.

8. Result Display and Testing: The analysis results, including the skill-match percentage, missing skills, and improvement suggestions, will be returned to the client interface. The application will be tested for calculation accuracy, database connectivity, input validation, and client handling.

System Architecture



Task Completed / Pending

Task	Status
Project idea and problem definition	Done
Project title and objectives finalisation	Done
Identification of target users and project requirements	Done
High-level system architecture design	Done
Frontend interface development	Done
Basic UI layout and page design	Done
Java backend development	Done
Java Socket Programming	Done
MySQL database design and connectivity	Pending
Career and required-skills dataset preparation	Pending
Skill-match percentage calculation	Done
Missing-skills identification module	Done
Personalised recommendation module	Pending
Multithreading implementation	Done
Frontend and backend integration	Done
Input validation and error handling	Pending
Functional and concurrent-user testing	Pending
Final integration, debugging and documentation	Pending

Challenges/Roadblocks

1. **Frontend Development Designing** a user-friendly interface to enter skills, select a target career, and display analysis results requires planning the layout and user interaction.

2. **Backend Development** The Java backend is yet to be developed. Connecting the frontend with the server and implementing the application logic will be a major upcoming task

3. **Client–Server Communication** Implementing Java Socket Programming to transfer student information between the client and server may require handling connection failures, request processing, and response management.

4. **Database Managemen**t Designing a MySQL database for career categories, required skills, and proficiency levels will require a suitable database structure and accurate skill-related information.

5. **Skill Gap Analysis** Developing a reliable method to compare students' existing skills with career requirements and calculate a meaningful skill-match percentage will require careful algorithm design.

6. **Recommendation Module** Creating effective recommendations based on the detected skill gap and proficiency analysis requires the appropriate logic tying the evaluation result to particular suggestions.

7. **Multithreading Implementation** Handling multiple client requests in a way that maintains reliable communication and server performance might be challenging.

8. **System Integration and Testing** Connecting the frontend, Java backend, database, and analysis modules into a coherent whole will involve performing integration testing, debugging, and validation.

   Current progress: The Frontend interface has been developed as well as integration work.

Project Outcome / Deliverables

1. Frontend User Interface

The frontend interface has been developed to provide a user-friendly platform for students to interact with the Skill Bridge system.

2. Career Selection Interface

The system will allow students to select their desired career path and enter their existing skills and proficiency levels.

3. Java Backend System

A Java-based backend will be developed to process user requests and manage communication between the frontend, database, and analysis modules.

4. Client–Server Communication

Java Socket Programming will be implemented for this task to ensure effictive data exchange between the client and server.

5. Career and Skills Database

The MySQL database will be created and populated with information regarding career categories, required skills, and the necessary proficiency levels.

6. Skill Gap Analysis Module

The system will compare students existing skills with the requirements of their selected careers and identify missing or insufficient skills.

7. Skill Match Percentage

The system will analyze and display to the user a percentage based on the students skills and the required skills of the career selected by the student.

8. Personalised Recommendation Module

The system will suggest possible ways to improve skills not present in student or which need improvement and recommend resources for the said learning .

9. Multithreading Support

Java multithreading concept will be used to attend to multiple client at a time.

10. Integrated and Tested Application

The final deliverable will be an integrated application combining the frontend, Java backend, MySQL database, skill-gap analysis, and recommendation modules.

Current outcome: The frontend interface has been completed. The remaining deliverables are planned for development and validation during Phase 2

Progress Overview

The Skill Bridge: Smart Career and Skill Gap Analysis System has progressed from project planning to the frontend development stages. The project objectives, requirements, and high-level system architecture have been established. Additionally, the frontend interface has been created, while the remaining steps will involve backend development, database connection, skill-gap analysis, and recommendations.

Project Phases

Phase 1 – Project Planning
In the first phase, the project idea, problem statement, objectives, and scope were established. The main objective of the project is to bridge the gap between the skills that a student has and the skills that they need to pursue their desired career.

Phase 2 – System Design and Architecture
The second phase is related to developing the system's architecture and designing its elements. The project's architecture consists of the client interface, the Java server, MySQL database, the skill gap detection module, and the recommendation module.

Phase 3 – Frontend Development
The project's frontend development has been completed and is used to present a user-friendly layout for the Skill Bridge: Smart Career and Skill Gap Analysis System application. The frontend will be the first point of interaction for students using the application.

Phase 4 – Backend Development (Done)
The Java backend will be developed to handle all the requests from the frontend. It will also include the application's logic to ensure smooth flow between the backend and the database. This component will connect the application to the database and the skill-gap analysis and recommendation modules.

Phase 5 – Client–Server Communication (Done)
Java Socket Programming will be used to facilitate communication between the client and the server to enable the exchange of student information and analysis results.

Phase 6 – Database Development (Pending)
This component will be designed to store all the career categories, skills, and proficiency levels that the application will use to analyze student skills.

Phase 7 – Skill Gap Analysis and Recommendations (Pending)
The analysis component will be developed to facilitate the comparison of student skills against the skills required in different careers to determine the gap, percentage of matching skills, and areas of improvement.

Phase 8 – Multithreading and Final Integration (Pending)
Multiple requests in an application require the use of multithreading. The backend will be connected to the frontend, database, and analysis module. Finally, the system will be tested and tuned to ensure it meets the quality standards.

Codebase Information

Project Title: Skill Bridge – Smart Career & Skill Gap Analysis System

Programming Language: Java (planned for backend development)

Frontend Status: Completed

Backend Status: Pending development

Main File

```
SkillBridge/
│
├── frontend/
│   └── User Interface
│
├── src/
│   ├── client/
│   │   └── Client.java
│   │
│   ├── server/
│   │   └── Server.java
│   │
│   ├── analysis/
│   │   └── SkillGapAnalyzer.java
│   │
│   ├── database/
│   │   └── DatabaseConnection.java
│   │
│   └── recommendation/
│       └── RecommendationEngine.java
│
├── database/
│   └── skillbridge.sql
│
├── tests/
│
└── README.md
```

Technologies To Be Used Or To Be Planned

- **Frontend Developed:** the specific framework or language depends on the actual implementation.
- **Java:** To perform planned for backend logic and application processing.
- **Java Socket Programming:** Planned for client-server communication.
- **MySQL:** To store and retrieve the skill information.
- **Java Multithreading:** To process multiple clients at a time.

Testing and Validation Status

Test Area	Status	Description
Frontend interface review	Done	Review the layout, navigation, and UI elements.
Input validation	Done	Verify that user inputs are entered correctly.
Backend functionality testing	Pending	Test Java server-side processing.
Client-server communication	Pending	Verify data exchange using Java Socket Programming.
Database validation	Pending	Check MySQL connectivity and career-skill data retrieval.
Skill-match calculation	Done	Validate match percentages and missing-skill identification.
Recommendation testing	Pending	Verify suggestions generated for identified skill gaps.

Deliverables Progress

Progress Summary

Completed

- Frontend interface development.
- Initial project planning and requirements definition.

In Progress / To Be Verified

- System architecture design and documentation.

Pending

- Backend, database, skill-gap analysis, recommendations, multithreading, integration, and comprehensive testing.

Expected Final Deliverables

The final project is expected to provide a working Skill Bridge application that allows students to select a career, compare their skills with career requirements, view a skill-match percentage, identify missing skills and receive recommendations for improvement.